Assignment → 1

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Define Transducer →

→ A transducer is a device the converts a physical quantity such as temperature, pressure displacement, force, speed, light or sound into another form of energy, usally an electrical signal or converts an electrical signal into a physical output.

In measurement and control system, a transducer acts as an interface between the physical world and the electronic system. It sense a physical quantity, converts it into a usable signals and allows the signal to be measured, displayed recarded or used for automatic control.

General working principle →

physical quantity → Sensing element → Conversion element → electrical/usable output

Example →

→ A thermocouple senses temperature and produce a small electrical voltage related to the temperature similarly a microphone converts sound vibrations into a electrical Signal.

⊕ Important functions of a transducer →

a) Sensors or detects a physical quantity.

b) Converts the measured quantity into a suitable signal.

c) Provides an output that can be measured, processed or cancelled.

d) forms an important between link sensors and electronics instrumentation system.

⊕ Classification of Transducer →

Transducer are classified as →

a) Based on Energy Source →

→ Active transducer:- Generate its own electrical output and does not require an external power supply.

ex→ Thermocouple, piezoelectric transducer.

→ Passive Transducer:- Requires an external power supply to operate.

ex→ Strain guage, RTD.

b) Based on Output Signal :-

→ Analog Transducer :- Produces a continuous output signal.

ex→ Thermister.

→ Digital Transducer :→ Produces a discrete / digital output signal.

ex→ Digital encoder.

c) Based on function →

→ Primary transducer :→ Directly sense the physical quantity.

ex→ Bourdon tube.

→ Secondry Transducer :→ Converts the output of a primary transducer into an electrical signal.

ex→ LVDT.

d) Based on Direction of Conversion →

→ Input Transducer :→ Converts a physical quantity into an electrical signal.

ex→ Microphone.

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→ Output Transducer:→ Converts an electrical signal into a physical part (output)

ex → Loudspeaker

c) Based on Principle of Operation →

→ Resistive :→ Works by change in resistance.
→ Inductive :→ Works by change in Inductance.
→ Capacitive :→ Works by change in capacitance.
→ Piezoelectroic :→ works on the piezoelectric effect
→ Photoelectric :→ works by converting light into an electrical signal.

3. Types of Transducer and their Characteristics →

a) Type → Resistive transducer
working principal → Input changes electrical resistance
ex → strain guage, RTD.

b) Type → Inductive transducer
working principal → Input changes inductive or magnetic reluctance
ex → LVDT.

c) Type → Capacitive transducer
working principle → Input change capacitance by changing distance
ex → characteristics → High sensitive, low power

d) Piezo electroric transducer
working principle → Mechanical stress on a pzm prod. elc charge
characteristics → High frequency response, compact

e) Type → Thermoelectric transducer
working principle → temperature difference produce
ex → Thermocouple
characteristics → self-generating, wide temperature

f) Type → Photoelectric transducer
working principle → light energy is converted into an electric signal.
ex → motion sensor
characteristics → fast response, non-contact measurement depends on light intensity.

g) Type → Hall-effect transducer
working principle → Magnetic field producer a hall voltage.
ex → Proximity sensors,
characteristics → Non-contact, useful for current, speed and position measurement.

h) Type → Ultrasonic transducer
working principle → uses ultrasonic waves for sensing distance
ex → Sonography
characteristics → Non-contact, useful for distance.

Important characteristics of a good transducer

a) Accuracy :→ Ability to give an output close to the true value.

b) Sensitivity :→ Ratio of change in output to the corresponding change in inoutput.

c) Linearity :→ Degree to which the output follows a straight line relationship with the input.

d) Resolution :→ Smallest change in input that can be detected by transducer.

e) Repeatability :→ Ability to produce nearly the same output reapeated measure ments under the same conditions.

f) Range :→ Minimum to maximum value of input that can be measured reliably.

. Ability to maintain its performance and calibration with time and changing conditions.

④ Overview of the Mechanical Actuation System :→

A mechanical actuation system is a sytem that produces and control mechanical motion or force by converting an input signal or energy into a mechanical output. It is commonly used in ~~mecha~~ machine, robots, automated manufacturing system and mechatronic products.

An actuators is the main element that creates physical movement. depending on the application, the actuators may produce linear motion,

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rotary motion, force, torque or controlled displacement.

⑧ Types of Mechanical Actuators :→

a) Electric actuators :→ Use electric energy to produce motion. common example are→ DC motors, AC motors, servo motors and stepper motion

b) Pneumatric actuators :→ Use compressed air to produce linear or rotary motion. They are clean and fast and are widely used in automation.

c) Hydroulic actuators :→ Use mechanical arrangements such as screw mechanisms, cams, gears, and linkages to converts or transmit motion.

d) Mechanical actuators :→ Use mechanical arrangements even as screw mechanisms, cams, gears and linkages to converts or transmit motion.

⑧ Working of a mechanical Actuation System.

a) A desired motion or position is specified by the user or control program.

b) The controller produces a suitable control signal.

c) The driver/power stage supplied appropriate power to the actuators.

d) The actuator converts the supplied energy into mechanical motion or force.

e)

e) The mechanical transmission transfer and modifies the motion according to the requirement.

f) The load recives the mechanical output.

g) If fudback is used, sensors measure the actual output and send it to the controller for correction.

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